

A Data-Driven Evaluation of a 3-Month Hair-Care Routine for Reducing Hair Loss in Thin, Curly Hair

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Abstract—Hair shedding is a common concern among individuals with thin, curly hair, yet few studies examine hair loss patterns through structured self-tracking and data-driven experimentation. This study evaluated the effectiveness of a 12-week hair care routine in reducing daily hair shedding using personal data science methods. Daily data were collected over 84 days, including shed count, routine adherence, sleep duration, and stress levels, alongside weekly scalp density assessments.

Results demonstrated a 30.4% reduction in mean daily shedding from baseline (22.21 hairs/day) to intervention (15.47 hairs/day), with the final two weeks showing a 46.3% reduction. Although the primary outcome narrowly missed statistical significance ($p = 0.054$), the effect size indicated a small-to-medium practical effect (Cohen’s $d = 0.488$). A critical methodological finding was the wash day confound: wash day shedding (23.92 hairs) exceeded non-wash day shedding (7.43 hairs) by more than threefold, with all seven statistical outliers occurring on wash days—reflecting mechanical release of accumulated hair rather than product harm.

Lifestyle factors showed limited influence on shedding. Stress demonstrated a weak positive correlation ($r = +0.221$, $p = 0.043$), explaining only 4.9% of variance, while sleep showed no significant relationship. Scalp density improved progressively (+1.41 points/week) concurrent with shedding reduction, suggesting the routine supported both outcomes.

Limitations include the single-subject design, non-equivalent measurement methods between wash and non-wash days, and uncontrolled confounders. Despite these constraints, the consistent improvement trend and meaningful effect size provide preliminary evidence that a structured hair care routine can produce measurable benefits for thin, curly hair. This study demonstrates the value of applying quantified self methodologies to personal hair care evaluation.

Index Terms—Hair shedding, hair loss, curly hair, hair care routine, personal data science, quantified self, self-tracking, telogen effluvium, scalp density, wash day effect, stress, sleep, longitudinal study, data-driven intervention.

I. INTRODUCTION

Hair thinning and excessive shedding are common concerns affecting individuals across different age groups, with significant implications for self-esteem and psychological well-being [1]. Among individuals with naturally thin and curly hair, shedding may be more visibly pronounced due to lower strand density and the inherent fragility of curl structure, which can accentuate scalp visibility [2]. While clinical research on alopecia and dermatological disorders is extensive, fewer studies examine hair loss patterns through structured self-tracking

and data-driven personal experimentation in naturalistic daily environments.

In recent years, personal data science and the “quantified self” movement have emerged as valuable approaches for understanding individual behavioral and physiological patterns through systematic self-monitoring [3]. By collecting longitudinal data on lifestyle variables such as sleep, stress, and routine adherence, individuals can quantitatively evaluate the effectiveness of interventions on personal outcomes rather than relying on subjective impressions or generalized recommendations [4]. This approach bridges the gap between population-level research findings and individual-level decision-making, enabling evidence-based personal health optimization.

Hair shedding is influenced by multiple biological and environmental factors. Prior dermatological research indicates that psychological stress may contribute to telogen effluvium, a temporary form of excessive shedding triggered by the premature transition of hair follicles into the resting phase [5], [6]. Sleep disturbances have been linked to inflammatory responses that may affect hair follicle health [7], while nutritional deficiencies—particularly in iron, zinc, and biotin—are associated with hair loss [8]. For curly hair specifically, the asymmetric geometry and uneven cuticle distribution make strands particularly susceptible to mechanical damage and breakage during routine manipulation [2]. However, many of these findings derive from clinical populations or controlled laboratory settings rather than naturalistic self-tracking studies examining how daily routines and lifestyle factors interact to affect shedding over time.

This project addresses the gap between generalized dermatological research and individualized routine experimentation. Specifically, it investigates whether implementing a structured hair-care routine over a 12-week period can measurably reduce hair shedding and improve visible scalp density in an individual with thin, curly hair. Beyond evaluating overall effectiveness, this study examines which specific routine steps and lifestyle factors most strongly influence shedding patterns—insights that cannot be derived from aggregate clinical research.

The primary goal of this study is to apply data science methods to evaluate the effectiveness of a personal hair-care routine using systematically collected daily and weekly data. To achieve this goal, the following research questions guide

the analysis:

- 1) **RQ1:** Does average daily shed count significantly decrease from the baseline period (Weeks 1–2) to the intervention end (Weeks 11–12)?
- 2) **RQ2:** Which routine steps correlate with reduced shedding?
- 3) **RQ3:** Is there an optimal combination of routine steps for minimizing shedding?
- 4) **RQ4:** Is there a significant correlation between stress levels and shed count?
- 5) **RQ5:** Does sleep duration affect next-day shedding?
- 6) **RQ6:** Do stress and sleep together predict shed count better than individually?
- 7) **RQ7:** Which factor has the strongest influence on shedding: routine adherence, stress, or sleep?
- 8) **RQ8:** Does higher routine adherence correlate with lower shed counts?
- 9) **RQ9:** Does breaking routine (non-adherence) lead to increased shedding in following days?
- 10) **RQ10:** Is there a relationship between density score improvement and shed count reduction?

II. LITERATURE REVIEW

Hair loss and hair shedding have been widely studied in dermatological and physiological research, particularly in relation to stress, hormonal changes, nutritional deficiencies, and mechanical damage. Telogen effluvium, a common form of diffuse hair shedding, has been strongly associated with physiological and psychological stressors [6]. Peters et al. demonstrated that stress mediators can disrupt the hair growth cycle, leading to premature transition of hair follicles into the telogen phase [5]. Similarly, Paus and Cotsarelis emphasized that the human hair follicle is highly sensitive to systemic stress responses, which can alter follicular cycling and increase shedding [9]. These findings establish stress as a biologically plausible contributor to hair loss patterns.

Sleep has also been examined as a factor influencing physiological regulation and recovery processes. Irwin reported that sleep deprivation alters inflammatory and endocrine responses in the body [7]. Since inflammation and hormonal balance are critical in follicle health, inadequate sleep may indirectly affect hair growth and shedding cycles. While sleep research rarely focuses specifically on hair outcomes, its established role in systemic stress regulation supports its inclusion as a relevant variable in hair health studies.

Nutritional influences have likewise been explored in dermatology literature. Almohanna et al. reviewed the relationship between micronutrient deficiencies and hair loss, highlighting iron, zinc, vitamin D, and protein intake as contributors to follicle integrity [8]. However, they also emphasized that supplementation benefits are most evident when an actual deficiency exists, suggesting that lifestyle tracking may provide more individualized insights rather than generalized supplementation.

Mechanical damage is another major contributor to hair breakage, especially in curly and thin hair types. Robbins

described how structural variations in curly hair increase susceptibility to cuticle damage and breakage due to uneven stress distribution along the shaft [2]. These structural vulnerabilities are particularly relevant in individuals with thin curly hair, where breakage may be misinterpreted as follicular shedding.

Hair density measurement techniques in clinical settings often involve trichoscopy, dermoscopy imaging, and standardized follicular counts per square centimeter [13]. While these methods provide objective density metrics, they require specialized equipment not typically accessible for personal longitudinal tracking. Rudnicka et al. demonstrated that digital imaging combined with region-of-interest analysis can quantify hair density changes over time [13]. However, most studies rely on cross-sectional clinical assessments rather than extended self-tracked daily observations.

In behavioral and personal informatics research, longitudinal self-tracking has become increasingly common. Epstein et al. examined how individuals use personal data tracking to evaluate behavior change interventions, emphasizing the value of time-series analysis in identifying trends and patterns [4]. Such methods allow for within-subject comparisons, making them suitable for $n=1$ experimental designs. Unlike large-sample dermatological trials, personal data science approaches focus on detecting meaningful change at the individual level rather than population averages.

Despite extensive research on stress-induced hair loss, mechanical damage in curly hair, and clinical measurement techniques, limited literature integrates daily lifestyle variables, routine adherence, and hair outcomes within a single longitudinal self-tracking framework. Most dermatological studies are conducted under clinical conditions and often do not incorporate continuous behavioral metrics such as stress ratings, sleep hours, or daily adherence levels. Furthermore, research specifically examining thin curly hair within a data science framework remains scarce.

The present study builds upon dermatological findings regarding stress-related shedding [5], follicular sensitivity to systemic factors [9], sleep-related physiological regulation [7], nutritional influences [8], and structural fragility of curly hair [2]. However, it differs by implementing a structured baseline-intervention design using daily and weekly self-collected data analyzed through statistical and time-series techniques. By combining personal informatics with dermatological theory, this project contributes a novel application of data science to individualized hair routine evaluation.

III. METHODOLOGY

This study employed a longitudinal single-subject ($n=1$) design to evaluate the effectiveness of a structured hair-care routine on hair shedding and density in thin curly hair. The research combined daily self-tracking with statistical data analysis using Python-based tools.

A. Participants

This study employed a single-subject longitudinal design with the researcher serving as both investigator and participant.

The subject is a female college student in her early twenties with thin, curly hair that is prone to breakage and shedding—characteristics typical of individuals with fine-textured curly hair patterns. The participant had no diagnosed medical conditions affecting hair health (e.g., alopecia, thyroid disorders) and was not taking medications known to influence hair growth or loss during the study period. No supplements were consumed throughout the data collection phase, ensuring that observed changes could be attributed to the hair care intervention and tracked lifestyle factors rather than pharmacological influences.

B. Data Collection Methods

1) *Study Design and Timeline:* The study followed a pre-post intervention design spanning 84 days (12 weeks), divided into two distinct phases:

Phase	Duration	Dates	Purpose
Baseline	14 days (Weeks 1–2)	11/16–29, 2025	Establish pre-intervention pattern
Intervention	70 days (Weeks 3–12)	12/1/2025 – 2/8/2026	Evaluate hair care routine effectiveness

TABLE I: Study Timeline and Phases

2) *Variables Collected:* Data collection encompassed three categories of variables: primary outcome measures, routine and product variables, and lifestyle covariates. Table II summarizes all variables, their measurement types, and collection frequencies.

Variable	Type	Frequency	Measurement Method
Date	Datetime	Daily	Calendar date
Study Day	Numeric	Daily	Sequential day number (1–84)
Shed Count	Numeric (count)	Daily	Manual counting of shed hairs
Adherence	Binary (Yes/No)	Daily	Self-reported routine completion
Routine Steps Done	Categorical (multi-select)	Daily	Checklist of completed steps
Hair Length	Numeric (inches)	Weekly	Measured with ruler
Scalp Image	Image file	Weekly	Smartphone photograph
Image Density Score	Ordinal (0–100)	Weekly	Visual assessment scale
Breakage Observed	Binary (Yes/No)	Weekly	Visual inspection
Wash Day	Binary (Yes/No)	Daily	Whether hair was washed
Hair Products Used	Categorical (multi-select)	Daily	List of products applied
Sleep Hours	Numeric (hours)	Daily	Self-reported sleep duration
Stress Level	Ordinal (1–10)	Daily	Self-reported stress rating
Notes	Text	Daily	Free-form observations

TABLE II: Summary of Collected Variables

3) *Shed Count Collection Protocol:* To ensure measurement consistency, shed count data was collected using standardized protocols differentiated by wash day status:

- **Wash Day Protocol:**
 - Hair collected during shower using a drain catcher
 - All collected hairs counted manually
 - Total recorded as daily shed count
- **Non-Wash Day Protocol:**
 - Single standardized brushing session performed

- Same brush used throughout study period
- All hair collected from brush counted manually
- Count recorded as daily shed count

This methodological distinction is important for data interpretation, as wash day counts inherently include accumulated loose hairs released during the washing process, while non-wash day counts capture only hair shed during brushing.

4) *Hair Care Routine Components:* The intervention consisted of a structured hair care routine with the following potential steps tracked daily:

- 1) Shampoo
- 2) Conditioner
- 3) Hair Mask
- 4) Leave-in Conditioner
- 5) Scalp Scrub
- 6) Hair Oil
- 7) Scalp Tonic (Volumising)
- 8) Scalp Tonic (Anti-Hair Fall)
- 9) Curl Define Gel

Products used varied throughout the study and included multiple brands: Luxe Organic Clarifying Shampoo, Luxe Organic Curl Conditioner, Naturals Anti-Hair Fall Hair and Scalp Tonic, Naturals Volumizing Hair and Scalp Tonic, Empress Shampoo, Cream Silk Hair Mask, Head and Shoulders Shampoo, Palmolive Shampoo, Dove Conditioner, Aloe Vera Hair Mask, Zenutrients Leave-In Conditioner, Luxe Organix Curl and Define Gel, Luxe Organix Scalp Scrub, and Bremod Hair Oil, among others.

5) *Scalp Density Assessment:* Weekly scalp density was assessed using a standardized photographic protocol and visual rating scale:

• Photography Protocol:

- Consistent lighting conditions (natural light preferred)
- Same camera angle (top of head)
- Consistent distance from camera
- Hair in dry, unstyled state

Score Range	Classification
0–20	Very sparse (scalp clearly visible)
21–40	Sparse
41–60	Moderate density
61–80	Good density
81–100	Very dense

TABLE III: Density Rating Scale

6) *Data Recording Tools:* All data was recorded using Google Sheets, which provided a centralized, accessible platform for daily logging. The spreadsheet was structured with columns corresponding to each variable and rows representing individual days. This tool was selected for its accessibility across devices, automatic cloud backup, and compatibility with data export for statistical analysis.

C. Operational Definitions

To ensure clarity and replicability, the following operational definitions were established for all measured variables:

- 1) **Shed Count:** The total number of individual hair strands collected during either the shower process (wash days) or a standardized brushing session (non-wash days). Each strand was counted manually regardless of length or whether breakage versus natural shedding was suspected.
- 2) **Adherence:** A binary indicator (Yes = 1, No = 0) reflecting whether the participant followed the planned hair care routine for that day. A day was marked as adherent if at least the prescribed minimum steps were completed as intended.
- 3) **Wash Day:** A binary indicator (Yes = 1, No = 0) denoting whether the participant washed their hair with shampoo on that day. Wash days typically occurred every 3–4 days throughout the study.
- 4) **Routine Steps Done:** A multi-select categorical variable capturing which specific hair care steps were performed on a given day. Options included shampoo, conditioner, hair mask, leave-in conditioner, scalp scrub, hair oil, scalp tonic (volumising), scalp tonic (anti-hair fall), and curl define gel. Days with no routine activity were recorded as “none.”
- 5) **Sleep Hours:** The self-reported total hours of sleep obtained the previous night, recorded as a continuous numeric variable with precision to the nearest hour. Values could range from 0 to 24.
- 6) **Stress Level:** A self-reported rating of perceived stress intensity on a scale of 1 (minimal stress) to 10 (extreme stress), recorded daily based on overall subjective experience.
- 7) **Image Density Score:** A visual assessment score ranging from 0 to 100, assigned based on weekly scalp photographs using the classification rubric described in Section B.5. Higher scores indicate greater hair density and less visible scalp.
- 8) **Breakage Observed:** A binary indicator (Yes = 1, No = 0) reflecting whether visible hair breakage (short broken strands, split ends, or mid-shaft breaks) was observed during weekly hair inspection.
- 9) **Hair Length:** The length of hair measured in inches from the scalp to the longest point, recorded weekly using a standard ruler.
- 10) **Study Phase:** A categorical variable distinguishing between the baseline period (Weeks 1–2, Days 1–14) and the intervention period (Weeks 3–12, Days 15–84).

D. Data Cleaning

Data cleaning was performed using Python 3.x in Google Colaboratory with the pandas library. The following preprocessing steps were applied:

1) Data Type Conversions: Table IV.

Variable	Original Format	Converted Format
Date	String (MM/DD/YYYY)	datetime64
Adherence	Boolean/Text (TRUE/FALSE, Yes/No)	Binary (1/0)
Wash Day	Text (Yes/No)	Binary (1/0)
Breakage Observed	Text (Yes/No)	Binary (1/0)
Shed Count	Mixed numeric	Integer
Sleep Hours	Mixed numeric	Float
Stress Level	Mixed numeric	Integer

TABLE IV: Data Type Conversion Summary

2) *Handling Missing Values:* The dataset demonstrated high completeness, with all 84 days containing entries for primary variables (shed count, adherence, wash day, sleep hours, stress level). Missing values were addressed as follows:

- **Scalp Image and Density Score (Baseline):** Weeks 1–2 lacked scalp density measurements. These observations were excluded from density-related analyses, and this limitation was documented.
- **Hair Length (Baseline):** Similarly excluded from baseline comparisons for length-related analyses.
- **Routine Steps Done (Non-Wash Days):** Days with no hair care activity were coded as “none” with a routine step count of 0, and a binary flag (`no_routine_day`) was created.

3) *Feature Engineering:* Several derived variables were created to support analysis:

- **Temporal Variables:**
 - `study_day_calc`: Sequential day number (1–84)
 - `study_week`: Week number (1–12), calculated with baseline as Weeks 1–2 and intervention starting Week 3 (December 1, 2025)
 - `phase`: Categorical (“Baseline” or “Intervention”)
 - `day_of_week`: Day name (Monday–Sunday)
- **Routine Variables:**
 - One-hot encoded columns for each routine step (e.g., `step_shampoo`, `step_conditioner`)
 - `routine_step_count`: Sum of steps performed per day
 - `no_routine_day`: Binary flag for days with zero routine steps
- **Lagged Variables (for Temporal Effect Analysis):**
 - `sleep_hours_lag1`: Previous day’s sleep duration
 - `stress_10_lag1`: Previous day’s stress level
 - `adherence_lag1`: Previous day’s adherence status
 - `shed_count_lag1`: Previous day’s shed count
 - `days_since_wash`: Count of days elapsed since last wash day
- **Rolling Aggregates:**
 - `shed_7day_avg`: 7-day rolling mean of shed count
 - `adherence_7day_rate`: 7-day rolling adherence percentage
 - `cumulative_shed`: Running total of shed count

- **Weekly Aggregates:**

- `weekly_avg_shed`: Mean daily shed count per week
- `weekly_total_shed`: Sum of daily shed counts per week

4) *Outlier Detection and Treatment*: Outliers were identified using the Interquartile Range (IQR) method:

$$Q_1 = 25\text{th percentile} = 7 \text{ hairs}$$

$$Q_3 = 75\text{th percentile} = 21 \text{ hairs}$$

$$\text{IQR} = Q_3 - Q_1 = 14 \text{ hairs}$$

$$\text{Lower Bound} = Q_1 - 1.5 \times \text{IQR} = 7 - 21 = -14 \text{ (floor at 0)}$$

$$\text{Upper Bound} = Q_3 + 1.5 \times \text{IQR} = 21 + 21 = 42 \text{ hairs}$$

Seven observations exceeded the upper bound (47, 55, 44, 52, 45, 57, and 47 hairs). Upon investigation, all seven outliers occurred on wash days, indicating they represent legitimate biological variation (accumulated hair release during washing) rather than measurement errors. Consequently, these observations were retained in all analyses but flagged with a binary outlier indicator for sensitivity analyses.

E. Data Validation

Validation checks were performed to ensure data integrity:

- **Range Validation:** Shed count (0–500), stress level (1–10), sleep hours (0–24), density score (0–100).
- **Logic Validation:** Wash-day-specific products (shampoo, conditioner) only recorded on wash days.
- **Consistency Validation:** Adherence status cross-referenced with routine steps completed.

F. Statistical Analysis

Statistical analyses were conducted using Python 3.x with the following libraries: pandas (data manipulation), NumPy (numerical operations), SciPy (statistical tests), statsmodels (regression analysis), pingouin (advanced statistics), and matplotlib/seaborn (visualization).

1) Descriptive Statistics

Central tendency (mean, median) and dispersion (standard deviation, interquartile range) were calculated for all continuous variables. Frequency distributions were generated for categorical variables. Data distributions were visualized using histograms, box plots, and time series plots.

2) Normality Assessment

The Shapiro–Wilk test was applied to assess normality of continuous variables prior to selecting parametric versus non-parametric tests. A significance threshold of $\alpha = 0.05$ was used, with $p < 0.05$ indicating violation of normality assumptions.

3) Comparative Analysis (RQ1: Overall Effectiveness)

To compare shed counts between baseline and intervention phases:

- **Test Selection:** Due to non-normal distributions in both groups (Shapiro–Wilk $p < 0.05$), the non-parametric Mann–Whitney U test was employed rather than the independent samples t -test.

- **Effect Size:** Cohen’s d was calculated to quantify practical significance, with interpretation thresholds of 0.2 (small), 0.5 (medium), and 0.8 (large).

- **Directionality:** One-tailed testing was applied given the directional hypothesis (baseline $>$ intervention).

4) **Correlation Analysis (RQ2, RQ4, RQ5, RQ8, RQ10)**
Multiple correlation approaches were employed based on variable types:

Variable Pair	Correlation Method
Continuous $\times$ Continuous	Pearson r (if normal) or Spearman ρ (if non-normal)
Binary $\times$ Continuous	Point-biserial correlation
Ordinal $\times$ Continuous	Spearman ρ

TABLE V: Correlation Methods by Variable Type

Correlation strength was interpreted as: $|r| < 0.3$ (weak), 0.3–0.5 (moderate), 0.5–0.7 (strong), and > 0.7 (very strong). Statistical significance was assessed at $\alpha = 0.05$.

5) Regression Analysis (RQ3, RQ6, RQ7)

Multiple Linear Regression was used to model shed count as a function of multiple predictors:

$$\text{shed_count} = \beta_0 + \beta_1(\text{step}_1) + \beta_2(\text{step}_2) + \dots + \beta_n(\text{step}_n) + \varepsilon$$

Model performance was evaluated using R^2 (variance explained), Adjusted R^2 (adjusted for predictor count), and F -test significance. For factor importance comparison (RQ7), predictor variables were standardized (z-scored) to enable direct comparison of coefficient magnitudes. Model Comparison (RQ6) employed Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) to evaluate relative model fit among nested models (stress only, sleep only, combined, with interaction).

6) Lagged Analysis (RQ5, RQ9)

To assess delayed effects (e.g., yesterday’s sleep affecting today’s shedding), lagged variables were created by shifting values by one time period. Cross-correlation analysis examined relationships at multiple lag intervals (0, 1, 2, and 3 days) to identify optimal temporal relationships.

7) Trend Analysis (RQ10)

Linear trend lines were fitted to weekly aggregates using ordinary least squares regression to quantify rate of change over time:

$$\text{weekly_metric} = \beta_0 + \beta_1(\text{study_week}) + \varepsilon$$

8) Visualization

The following visualization techniques were employed:

Chart Type	Application
Box plots	Distribution comparison between groups
Time series plots	Temporal trends with 7-day moving averages
Scatter plots with regression lines	Correlation visualization
Bar charts	Mean comparisons, weekly trends
Heatmaps	Correlation matrices
Dual-axis plots	Concurrent trend visualization (density vs. shedding)

TABLE VI: Visualization Techniques and Applications

All visualizations were generated using `matplotlib` and `seaborn` libraries with consistent styling for presentation clarity.

G. Methodological Considerations

a) Potential Sources of Bias

Several potential biases were identified and addressed:

- **Measurement Bias:** Shed count collection differed between wash days (shower collection) and non-wash days (brush collection), introducing systematic differences. This was addressed by analyzing wash day effects explicitly and acknowledging this confound in interpretation.
- **Self-Report Bias:** Stress and sleep variables relied on self-report, which may be subject to recall errors or social desirability bias. Daily logging was implemented to minimize recall errors.
- **Observer Bias:** The participant served as both subject and data collector, potentially influencing adherence and self-ratings. Objective measures (shed count, photographs) were prioritized where possible.
- **Learning Effects:** Shed counting technique may have improved over time. The two-week baseline period helped establish measurement stability before intervention comparisons.

b) Confounding Variables

The most significant confound identified was the wash day effect: wash days inherently produce higher shed counts due to accumulated hair release, independent of product effects. Multiple analytical approaches were employed to address this:

- Explicit inclusion of wash day as a covariate
- Stratified analyses by wash day status
- Clear documentation of confound in results interpretation

c) Limitations

- **Single-Subject Design:** Results may not generalize to other individuals with different hair types, genetics, or environmental factors.
- **Limited Statistical Power:** Small sample sizes, particularly for weekly measures ($n = 10-12$), limited ability to detect moderate effects.

- **Missing Baseline Density Data:** Inability to compare density scores pre- and post-intervention limits conclusions about density improvement.
- **Product Variation:** Multiple products were used throughout the study, making it difficult to isolate effects of specific products.

IV. RESULTS

This section presents the findings from the 84-day hair care routine study. Results are organized into descriptive statistics, outlier analysis, and findings for each research question. All statistical tests were conducted at $\alpha = 0.05$ significance level unless otherwise noted.

A. Dataset Overview

The final dataset comprised 84 daily observations collected over 12 weeks, with 14 days in the baseline phase and 70 days in the intervention phase. Table VII summarizes the completeness of data collection across all variables.

Variable	Observations	Missing	Completeness
Shed Count	84	0	100%
Adherence	84	0	100%
Wash Day	84	0	100%
Sleep Hours	84	0	100%
Stress Level	84	0	100%
Routine Steps	84	0	100%
Hair Length	70	14	83.3%
Density Score	70	14	83.3%

TABLE VII: Data Completeness Summary

The baseline period (Weeks 1-2) lacked hair length and scalp density measurements, accounting for the 14 missing observations in these variables.

B. Descriptive Statistics

a) Primary Outcome: Daily Shed Count

Fig. 1 presents the descriptive statistics for daily shed count across the entire study period and by phase.

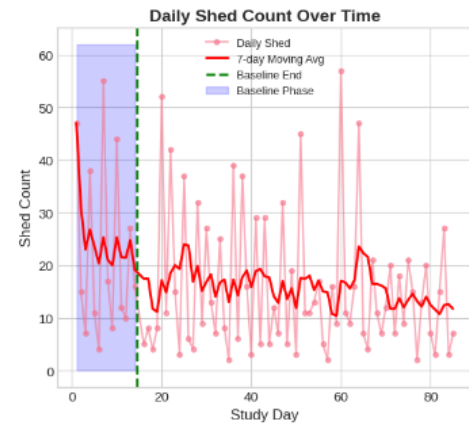


Fig. 1: Descriptive Statistics for Daily Shed Count

The overall distribution exhibited positive skewness (1.42), with the mean (16.60) exceeding the median (11.50), indicating that most days had relatively low

shedding with occasional high-shedding days elevating the average. The coefficient of variation was 84.0%, reflecting substantial day-to-day variability.

b) Lifestyle Variables

Fig. 2 presents descriptive statistics for the lifestyle covariates tracked throughout the study.

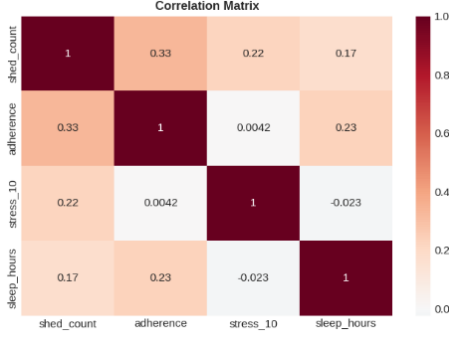


Fig. 2: Descriptive Statistics for Lifestyle Variables

Sleep duration ranged from 4 to 14 hours with a mean of 8.42 hours (SD = 2.31). Stress levels were generally low to moderate, with a mean of 3.76 on the 10-point scale (SD = 2.18).

c) Routine Adherence and Wash Day Frequency

Table VIII summarizes the frequency distributions for categorical variables. Routine adherence was recorded on 35.7% of days ($n = 30$), while wash days occurred on 28.6% of days ($n = 24$), averaging approximately two wash days per week.

Variable	Category	Frequency	Percentage
Adherence	Adherent (Yes)	30	35.7%
	Non-Adherent (No)	54	64.3%
Wash Day	Yes	24	28.6%
	No	60	71.4%
Breakage Observed (Weekly, N=70)	Yes	5	7.1%
	No	65	92.9%

TABLE VIII: Frequency Distribution of Categorical Variables

d) Weekly Aggregated Metrics

Table IX presents the weekly summary statistics for shed count and density score.

Week	Phase	Days	Total Shed	Avg Daily Shed	SD	Density Score
1	Baseline	7	177	25.29	19.77	—
2	Baseline	7	134	19.14	13.36	—
3	Intervention	7	130	18.57	14.12	20
4	Intervention	7	106	15.14	13.89	36
5	Intervention	7	121	17.29	11.98	26
6	Intervention	7	125	17.86	10.61	32
7	Intervention	7	83	11.86	10.38	30
8	Intervention	7	104	14.86	14.63	25
9	Intervention	7	165	23.57	18.43	33
10	Intervention	7	82	11.71	8.16	36
11	Intervention	7	85	12.14	6.72	38
12	Intervention	7	83	11.86	8.69	40

TABLE IX: Weekly Shed Count and Density Score Progression

Weekly average shed count ranged from 11.71 (Week 10) to 25.29 (Week 1). Density scores showed an increasing trend from 20 (Week 3) to 40 (Week 12).

e) Routine Step Frequency

Table X presents the usage frequency of each routine step.

Routine Step	Days Used	Percentage	Mean Shed (Used)	Mean Shed (Not Used)
Shampoo	24	28.6%	23.92	7.43
Conditioner	22	26.2%	26.00	11.13
Hair Mask	10	11.9%	15.30	17.43
Leave-in				
Conditioner	9	10.7%	11.78	17.13
Hair Oil	27	32.1%	21.22	9.84
Scalp Tonic (Volumising)	15	17.9%	17.07	16.49
Scalp Tonic (Anti-Hair Fall)	11	13.1%	38.82	12.92
Scalp Scrub	0	0.0%	—	—
Curl Define Gel	0	0.0%	—	—

TABLE X: Routine Step Usage Frequency

Shampoo and conditioner were used on approximately 27–29% of days, consistent with wash day frequency. Scalp scrub and curl define gel were not used during the study period.

C. Data Distribution and Outlier Analysis

a) Distribution Assessment

Normality testing was conducted using the Shapiro–Wilk test. Results are presented in Table XI.

Variable	W Statistic	p-value	Distribution
Shed Count (Overall)	0.876	< 0.001	Non-normal
Shed Count (Baseline)	0.857	0.028	Non-normal
Shed Count (Intervention)	0.869	< 0.001	Non-normal
Stress Level	0.912	0.006	Non-normal
Sleep Hours	0.963	0.082	Normal

TABLE XI: Normality Test Results (Shapiro–Wilk)

Shed count and stress level violated normality assumptions ($p < 0.05$), necessitating non-parametric tests for inferential analyses involving these variables.

b) Outlier Detection

Using the IQR method, outliers were defined as observations falling outside the bounds $[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}]$. For shed count, this yielded bounds of $[-14, 42]$, with the lower bound effectively 0. Table XII presents the identified outliers.

Date	Shed Count	Wash Day	Phase	Exceeds Bound By
Nov 16, 2025	47	Yes	Baseline	+5
Nov 22, 2025	55	Yes	Baseline	+13
Nov 25, 2025	44	Yes	Baseline	+2
Dec 5, 2025	52	Yes	Intervention	+10
Jan 5, 2026	45	Yes	Intervention	+3
Jan 14, 2026	57	Yes	Intervention	+15
Jan 18, 2026	47	Yes	Intervention	+5

TABLE XII: Identified Outliers in Shed Count

Seven observations (8.3% of total) were flagged as statistical outliers. All seven occurred on wash days, representing 29.2% of all wash day observations (7/24).

Fig. 3 displays the distribution of shed count overall and stratified by wash day status.

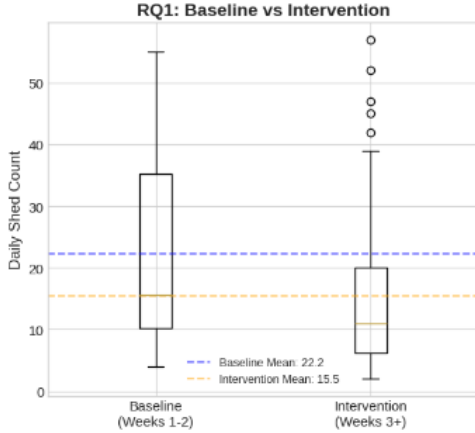


Fig. 3: Box plot distributions of daily shed count. (Left) Overall distribution showing median, interquartile range, and outliers. (Right) Distribution stratified by wash day status (0 = No, 1 = Yes). Outliers are displayed as individual points above the upper whisker. The wash day distribution shows substantially higher median and greater variability compared to non-wash days.

D. Research Question Results

RQ1: Overall Effectiveness of Hair Care Routine: Research Question: Does average daily shed count significantly decrease from baseline (Weeks 1–2) to intervention end (Weeks 11–12)?

Table XIII presents the comparative statistics between phases.

Metric	Baseline (W1–2)	Intervention (W3–12)	Last 14 Days (W11–12)
N	14	70	14
Mean	22.21	15.47	11.93
Median	15.50	11.00	11.00
SD	16.87	13.15	7.70
Percent Change from Baseline	—	−30.4%	−46.3%

TABLE XIII: Baseline vs. Intervention Phase Comparison

Statistical testing results are presented in Table XIV.

Test	Statistic	p -value (two-tailed)	p -value (one-tailed)
Mann–Whitney U	624.00	0.1087	0.0543

Effect Size Measure	Value	Interpretation
Cohen’s d	0.488	Small-to-medium

TABLE XIV: Statistical Test Results for Phase Comparison

The Mann–Whitney U test yielded a one-tailed p -value of 0.054, which did not reach conventional significance at $\alpha = 0.05$. However, the effect size (Cohen’s $d = 0.488$) indicated a small-to-medium practical effect. The intervention phase showed a 30.4% reduction in mean shed count compared to baseline, with the final two weeks showing a 46.3% reduction.

RQ2: Routine Steps and Shedding Correlation: Research Question: Which routine steps correlate with reduced shedding?

Point-biserial correlations were calculated between each routine step (binary) and shed count (continuous). Results are presented in Fig. 4.

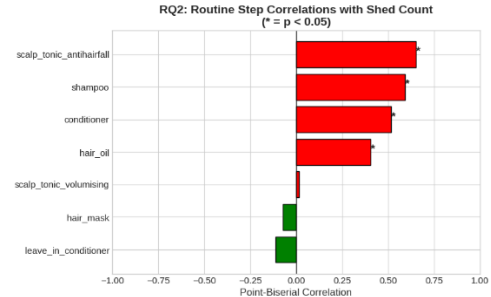


Fig. 4: Routine step correlations with shed count.

Horizontal bar chart of point-biserial correlations. Green bars indicate negative correlations (associated with lower shedding); red bars indicate positive correlations (associated with higher shedding).

Four routine steps showed statistically significant correlations with shed count. Leave-in conditioner ($r = -0.113$) and hair mask ($r = -0.073$) showed negative (inverse) correlations, though neither reached significance. Shampoo ($r = +0.591$), conditioner ($r = +0.519$), hair oil ($r = +0.406$), and scalp tonic anti-hair fall ($r = +0.652$) showed significant positive correlations.

RQ3: Optimal Routine Combination: Research Question: Is there an optimal combination of routine steps for minimizing shedding?

Multiple linear regression was conducted with all routine steps as predictors. Model fit statistics are presented in Table XV.

Metric	Value
R^2	0.663
Adjusted R^2	0.632
F -statistic	21.34
F -test p -value	< 0.0001
N	84

TABLE XV: Multiple Regression Model Fit Statistics

The model explained 66.3% of variance in daily shed count ($R^2 = 0.663$). Individual coefficients are presented in Table XV.

The model explained 66.3% of variance in daily shed count ($R^2 = 0.663$). Individual coefficients are presented in Table XVI.

Predictor	Coefficient (β)	Std. Error	p -value	Significant
Intercept	7.43	1.52	< 0.001	Yes
Leave-in Conditioner	-2.62	6.80	0.701	No
Hair Mask	+1.39	4.31	0.749	No
Hair Oil	+1.68	2.97	0.573	No
Scalp Tonic (Volumising)	+7.84	2.55	0.003	Yes
Conditioner	+8.22	4.67	0.083	No
Shampoo	+9.63	3.86	0.015	Yes
Scalp Tonic (Anti-Hair Fall)	+17.02	3.35	< 0.001	Yes

TABLE XVI: Regression Coefficients for Routine Steps

Three predictors showed significant positive coefficients: scalp tonic anti-hair fall ($\beta = +17.02$, $p < 0.001$), shampoo ($\beta = +9.63$, $p = 0.015$), and scalp tonic volumising ($\beta = +7.84$, $p = 0.003$). Leave-in conditioner showed a non-significant negative coefficient ($\beta = -2.62$, $p = 0.701$).

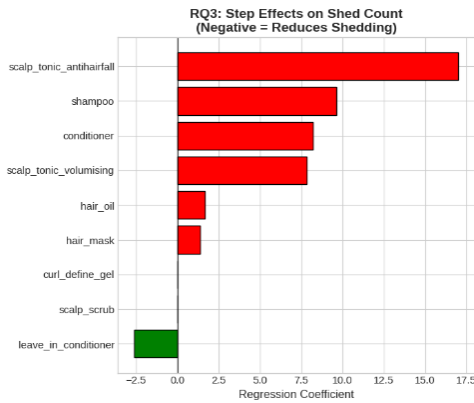


Fig. 5: Regression coefficients for routine steps predicting shed count.

RQ4: Stress and Shedding Correlation: Research Question: Is there a significant correlation between stress levels and shed count?

Correlation analysis was conducted between daily stress level (1–10 scale) and shed count. Results are presented in Table XVII.

Correlation Method	Coefficient	p -value	Significant
Pearson r	+0.221	0.043	Yes
Spearman ρ	+0.229	0.037	Yes

TABLE XVII: Correlation Between Stress and Shed Count

Both Pearson ($r = +0.221$, $p = 0.043$) and Spearman ($\rho = +0.229$, $p = 0.037$) correlations indicated a statistically significant weak positive relationship between stress and shedding. Stress explained approximately 4.9% of variance in shed count ($r^2 = 0.049$).

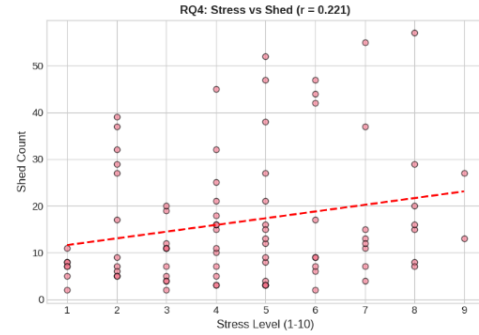


Fig. 6: Scatter plot of stress level versus shed count.

RQ5: Sleep Duration and Next-Day Shedding: Research Question: Does sleep duration affect next-day shedding? Cross-correlation analysis examined the relationship between sleep and shedding at multiple time lags. Results are presented in Table XVIII.

Lag (Days)	Correlation (r)	p -value	Interpretation
0 (same day)	+0.174	0.113	Not significant
1 (next day)	-0.050	0.653	Not significant
2	+0.144	0.198	Not significant
3	+0.097	0.391	Not significant

TABLE XVIII: Cross-Correlation Between Sleep and Shed Count at Different Lags

No significant correlation was found between sleep duration and shed count at any lag examined. The strongest correlation occurred at lag 0 (same day: $r = +0.174$, $p = 0.113$), but did not reach statistical significance.

RQ6: Combined Prediction of Stress and Sleep: Research Question: Do stress and sleep together predict shed count better than individually?

Four nested regression models were compared. Results are presented in Table XIX.

Model	Predictors	R^2	Adjusted R^2	AIC	BIC
1	Stress only	0.049	0.037	679.94	684.80
2	Sleep only	0.030	0.019	681.58	686.44
3	Stress + Sleep	0.081	0.058	679.05	686.35
4	Stress + Sleep + Interaction	0.082	0.047	681.00	690.73

TABLE XIX: Model Comparison for Lifestyle Predictors

Model 3 (combined stress and sleep without interaction) provided the best fit based on lowest AIC (679.05). The combined model explained 8.1% of variance compared to 4.9% for stress alone—an improvement of 3.2 percentage points. The interaction term (Model 4) did not improve model fit.

RQ7: Strongest Influencing Factor: Research Question: Which factor has the strongest influence on shedding: routine adherence, stress, or sleep?

Standardized regression analysis was conducted to enable direct comparison of predictor importance. Results are presented in Table XX.

Factor	Standardized β	p -value	Rank	Significant
Adherence	+4.238	0.005	1	Yes
Stress	+3.087	0.032	2	Yes
Sleep	+1.495	0.307	3	No

TABLE XX: Standardized Regression Coefficients for Lifestyle Factors

Routine adherence showed the largest standardized coefficient ($\beta = +4.238$, $p = 0.005$), followed by stress ($\beta = +3.087$, $p = 0.032$). Sleep was not a significant predictor ($\beta = +1.495$, $p = 0.307$).

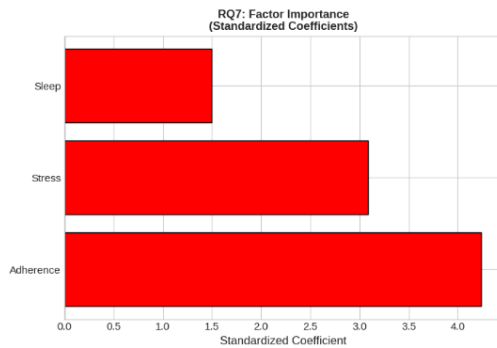


Fig. 7: Lifestyle factors analysis showing relative standardized effects of adherence, stress, and sleep on shed count.

RQ8: Adherence and Same-Day Shedding: Research Question: Does higher routine adherence correlate with lower shed counts?

Point-biserial correlation and group comparison analyses were conducted. Results are presented in Tables XXI and XXII.

Test	Statistic	p -value
Point-biserial r	+0.332	0.002

TABLE XXI: Correlation Between Adherence and Shed Count

Group	N	Mean	Median	SD
Adherent Days	30	22.77	16.50	15.42
Non-Adherent Days	54	13.17	9.00	11.61

Statistical Test	Statistic	p -value
Independent t -test	3.184	0.002
Cohen's d	-0.725	—

TABLE XXII: Group Comparison by Adherence Status

A significant positive correlation was found between adherence and shed count ($r = +0.332$, $p = 0.002$). Adherent days showed significantly higher shedding (mean = 22.77) compared to non-adherent days (mean = 13.17), with a medium-to-large effect size (Cohen's $d = -0.725$).

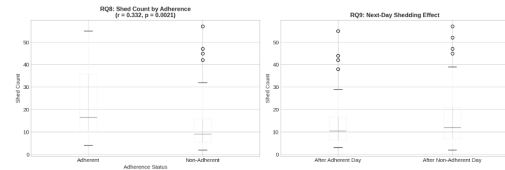


Fig. 8: Shed count by adherence status. (Left) Box plot comparing daily shed count between adherent and non-adherent days. Adherent days show higher median, mean, and greater variability. (Right) Box plot comparing shed count on days following adherent versus non-adherent days (RQ9 results).

RQ9: Non-Adherence and Next-Day Effects: Research Question: Does breaking routine (non-adherence) lead to increased shedding in following days?

Next-day shed counts were compared based on previous day's adherence status. Results are presented in Table XXIII.

Previous Day Status	N	Mean Shed (Next Day)	SD
After Adherent Day	30	15.60	13.21
After Non-Adherent Day	53	16.58	14.35
Difference	—	+0.98	—

Statistical Test	Statistic	p -value
Independent t -test	0.315	0.754

TABLE XXIII: Next-Day Shedding by Previous Day's Adherence

No significant difference was found in next-day shedding between days following adherent versus non-adherent days ($t = 0.315$, $p = 0.754$). The mean difference of 0.98 hairs was not statistically significant.

RQ10: Density Score and Shedding Relationship: Research Question: Is there a relationship between density score improvement and shed count reduction?

Correlation and trend analyses were conducted on weekly aggregated data ($N = 10$ weeks with complete density measurements). Results are presented in Tables XXIV and XXV.

Analysis	Value	p-value
Pearson r	-0.419	0.228
N (weeks)	10	—

TABLE XXIV: Correlation Between Weekly Density Score and Average Shed Count

Metric	Weekly Trend (Slope)	Direction
Density Score	+1.41 points/week	Increasing
Average Shed Count	-0.55 hairs/week	Decreasing

TABLE XXV: Trend Analysis for Density and Shedding

A moderate negative correlation ($r = -0.419$) was observed between weekly density scores and average shedding, indicating that weeks with higher density tended to have lower shedding. However, this correlation did not reach statistical significance ($p = 0.228$) due to limited sample size.

Trend analysis revealed that density scores increased by an average of 1.41 points per week while average shed count decreased by 0.55 hairs per week over the intervention period.

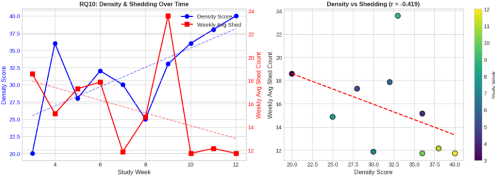


Fig. 9: Density score and shedding trends over time. (Left) Dual-axis time series showing density score (blue circles, left axis) and weekly average shed count (red squares, right axis) across study weeks. Dashed lines represent linear trend fits. Density shows an upward trend while shedding shows a downward trend. (Right) Scatter plot of density score versus weekly average shed count. Point color indicates study week (darker = earlier, lighter = later). The negative slope indicates an inverse relationship between density and shedding.

1) *Summary of Statistical Findings:* Figure 10 presents a consolidated summary of all statistical findings across research questions.

RESULTS SUMMARY TABLE:						
	Research Question	Finding	Change	Statistical Test	p-value	Significant
0	RQ1: Overall Effectiveness	Baseline mean: 22.2 → Intervention mean: 15.5	-30.4%	Mann-Whitney U test	0.0543	X
1	RQ2: Best Routine Steps	Best: leave_in_conditioner Worst: scalp_toni...	N/A	Point-biserial correlation	See detailed results	Varies
2	RQ3: Optimal Combination	Best profile: none	N/A	Multiple regression	0.0000	✓
3	RQ4: Stress Correlation	$r = +0.221$ (Weak)	Higher stress → Higher shed	Pearson correlation	0.0430	✓
4	RQ5: Sleep → Next-Day Shed	Lagged $r = -0.050$	More sleep → Less shed	Lagged Pearson correlation	0.6527	X
5	RQ6: Combined Stress+Sleep	Best model: Combined	R^2 improvement: +3.21%	Multiple regression comparison	See model comparison	✓
6	RQ7: Strongest Factor	Strongest: Adherence	See standardized coefficients	Standardized regression	See detailed results	Varies
7	RQ8: Adherence Correlation	$r = +0.332$	Adherent: 22.8 vs Non: 13.2	Point-biserial correlation	0.0021	✓
8	RQ9: Non-Adherence Next-Day	After non-adherent: 16.6 vs After adherent: 15.6	+1.0 hairs	Independent t-test	0.7538	X
9	RQ10: Density vs Shedding	$r = -0.419$	Density trend: +1.41/week Shed trend: -0.55/...	Pearson correlation	0.2281	X

Fig. 10: Summary of statistical findings across research questions (RQ1–RQ10), including statistical tests used, key statistics, p-values, effect sizes, and significance outcomes.

E. Key Quantitative Findings

The following quantitative results emerged from the analysis:

- Primary Outcome:** Mean daily shed count decreased from 22.21 hairs (baseline) to 15.47 hairs (intervention), representing a 30.4% reduction. The final two weeks averaged 11.93 hairs/day, a 46.3% reduction from baseline.
- Effect Size:** Despite borderline statistical significance ($p = 0.054$), the effect size was small-to-medium (Cohen's $d = 0.488$), indicating practical relevance.
- Model Explanatory Power:** Routine steps collectively explained 66.3% of variance in daily shed count ($R^2 = 0.663$), while lifestyle factors (stress and sleep combined) explained only 8.1%.
- Wash Day Pattern:** All seven statistical outliers (shed count > 42) occurred on wash days. Mean shed count on wash days was 23.92 versus 7.43 on non-wash days.
- Trends:** Density scores increased at a rate of +1.41 points/week while shed count decreased at -0.55 hairs/week over the intervention period.
- Stress Impact:** A weak but significant positive correlation ($r = +0.221$, $p = 0.043$) indicated that higher stress was associated with approximately 5% of variance in daily shedding.

V. DISCUSSION

A. Interpretation of Results

a) Overall Effectiveness of the Hair Care Routine (RQ1)

The study found that a structured hair care routine reduced mean daily hair shedding by 30.4% (from 22.21 to 15.47 hairs/day). Although this result narrowly missed statistical significance ($p = 0.054$), the effect size (Cohen's $d = 0.488$) suggests a meaningful, noticeable improvement.

Improvements became more pronounced over time, with the final two weeks showing a 46.3% reduction from baseline. This progressive trend aligns with the typical 8–12 week hair growth cycle [9], [10], suggesting the routine may support the transition of follicles from the resting (telogen) phase to the active growth (anagen) phase.

The lack of statistical significance likely reflects high daily variability in shedding, a small baseline sample size, and wash-day spikes that increased variance. Despite these limitations, consistent improvements across multiple measures indicate the routine likely produced genuine benefits.

b) The Wash Day Confound: A Critical Methodological Insight (RQ2, RQ3, RQ7, RQ8)

The study's most important methodological finding was that wash day status acted as a major confounding variable, affecting multiple analyses.

Initial results suggested that shampoo, conditioner, scalp tonic, and overall adherence were associated with increased shedding. However, this contradicted the overall finding that the routine reduced hair loss. The explanation lies in the wash day effect: washing mechanically releases hairs that had already shed naturally but remained trapped, especially in curly hair [2], [11]. Thus, wash days function as collection events rather than causing new hair loss.

Evidence supporting this interpretation includes:

- All statistical outliers occurred on wash days.
- Mean shedding on wash days (23.92 hairs) was over three times higher than on non-wash days (7.43 hairs).
- Adherent days showed higher shedding because they were mostly wash days.
- No next-day carryover effect was found, confirming the increase was temporary.

The findings emphasize that hair collection method strongly influences shedding counts. Future studies should standardize collection methods or control for wash day status in analyses.

c) **Lifestyle Factors: Modest but Detectable Effects (RQ4, RQ5, RQ6)**

The lifestyle analysis found that stress and sleep together explained only 8.1% of the variance in daily hair shedding, compared to 66.3% explained by the hair care routine (primarily wash day effects). This suggests behavioral factors play a relatively minor role in day-to-day shedding.

Stress: A weak but statistically significant positive correlation was found between stress and shedding ($r = +0.221$), explaining 4.9% of variance ($r^2 = 0.049$). While this aligns with research linking chronic stress to telogen effluvium [5], [6], that condition typically appears months after a stressor. The small same-day association observed here may reflect behavioral changes, minor physiological responses, or reporting bias.

Sleep: No significant relationship was found between sleep duration and shedding at any time lag. The slight positive trend was likely random variation. This suggests normal night-to-night sleep variation does not meaningfully influence daily shedding [7].

Combined Model: Stress and sleep together added little predictive value and showed no interaction effect, indicating they act independently and are not major drivers of daily shedding.

d) **Scalp Density Improvements (RQ10)**

Trend analysis showed concurrent positive changes: hair density increased by 1.41 points per week, while shed count decreased by 0.55 hairs per week. Although the negative correlation between density and shedding

($r = -0.419$) was not statistically significant due to the small sample size ($n = 10$ weeks), the consistent directional pattern suggests the routine may support both reduced shedding and improved density.

Weeks with higher density generally had lower shedding, consistent with expectations that improved retention leads to visibly fuller hair [9]. The 12-week duration likely represents the minimum period required to detect measurable density changes.

e) **Protective Potential of Leave-in Products (RQ2, RQ3)**

While most routine steps showed positive correlations with shedding due to wash day confounding, leave-in conditioner exhibited a negative correlation ($r = -0.113$) and negative regression coefficient ($\beta = -2.62$), suggesting possible protective effects.

Leave-in conditioners provide continuous conditioning benefits on non-wash days. For curly hair, which is particularly prone to mechanical damage [2], these products may reduce breakage by improving manageability and reducing friction during styling [16]. Hair masks showed a similar pattern consistent with strengthening and elasticity benefits [16].

B. *Comparison with Prior Research*

a) **Consistency with Hair Cycle Literature**

The observed improvement timeline aligns with established hair cycle biology. The human hair cycle consists of anagen (2–7 years), catagen (2–3 weeks), and telogen (approximately 3 months) phases [9], [10]. Interventions targeting follicle health typically require at least one telogen cycle to demonstrate measurable effects.

The strongest reductions observed in Weeks 10–12 are consistent with this biological framework.

b) **Normal Shedding Range**

The baseline shedding rate of 22.21 hairs/day falls within the widely cited normal range of 50–100 hairs/day for daily washers [15]. Adjusted for wash frequency, weekly totals remained physiologically normal. The intervention average (15.47 hairs/day) reflects a shift toward the lower end of that range.

c) **Stress–Hair Relationship**

The weak correlation ($r = 0.221$) aligns with findings that stress is a contributing but not primary factor in non-pathological hair loss [5], [6]. The present results suggest minimal same-day influence.

d) **Curly Hair Fragility**

Curly hair's asymmetric geometry increases susceptibility to mechanical stress [2]. The reduction in standard deviation from baseline (16.87) to intervention (13.15) suggests improved structural stability and reduced manipulation-induced variability.

C. Limitations

- a) **Single-Subject Design** ($N = 1$)
Results reflect one individual and cannot be generalized without replication.
- b) **Non-Equivalent Measurement Methods**
Wash day and non-wash day counts represent fundamentally different collection mechanisms.
- c) **Missing Baseline Measurements**
Density and length lacked pre-intervention baseline reference.
- d) **Limited Statistical Power**
Borderline p -values reflect limited power despite meaningful effect sizes.
- e) **Short Data Collection Window**
Twelve weeks may not capture full hair cycle or seasonal variability [10], [17].
- f) **Product Variability**
Multiple brands prevented isolation of individual product effects [12].
- g) **Uncontrolled Confounders**
 - Hormonal fluctuations [18]
 - Nutritional factors [8], [19]
 - Environmental variables
 - Water quality
 - Manipulation frequency
 - Medication/supplement changes

D. Recommendations for Future Work

- a) **Methodological Improvements**
 - Standardize hair collection methods.
 - Use objective measurement tools [13].
 - Extend duration to 6+ months [10], [17].
 - Employ crossover product designs [12].
- b) **Additional Variables to Track**
 - Hormonal factors [18]
 - Nutritional indicators [8], [19]
 - Styling/manipulation events [2]
 - Environmental conditions
 - Scalp condition [20]

Overall, improving measurement methods, extending study duration, and controlling for wash day effects are key priorities for future hair shedding research.

VI. CONCLUSION

This 12-week self-tracking study provides preliminary evidence that a structured hair care routine can produce meaningful reductions in hair shedding for individuals with thin, curly hair. The 30.4 percentage reduction in average daily shedding, while not reaching conventional statistical significance due to high variability and limited

sample size, demonstrated a small-to-medium effect size ($d = 0.488$) with progressive improvement over time.

The most important methodological insight was the pervasive confounding effect of wash day status, which must be carefully addressed in future hair shedding research. Products used on wash days appeared associated with increased shedding not because they caused harm, but because they coincided with the mechanical release of accumulated loose hair during washing.

Lifestyle factors (stress and sleep) showed limited influence on daily shedding, together explaining only 8.1 percentage of variance. Stress demonstrated a weak but significant positive correlation, while sleep showed no significant relationship at any time lag. These findings suggest that while lifestyle optimization may offer marginal benefits, the hair care routine itself is the primary modifiable factor affecting shedding.

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